

Replace `detector-id` with the detector ID of the member account for which you will delete the threat entity list, and other placeholder values that are *shown in red*.

To delete threat IP list

1. Run [DeleteThreatIntelSet](#). Make sure to provide the `detectorId` of the member account for which you want to delete this threat IP address list. To find the `detectorId` for your account and current Region, see the **Settings** page in the <https://console.aws.amazon.com/guardduty/> console, or run the [ListDetectors](#) API.
2. Alternatively, you can do this by running the following AWS Command Line Interface command and make sure to replace the `detector-id` with the detector ID of the member account for which you will delete the threat IP list.

```
aws guardduty delete-threat-intel-set \  
--detector-id 12abc34d567e8fa901bc2d34e56789f0 \  
--threat-intel-set-id d4b94fc952d6912b8f3060768example
```

Replace `detector-id` with the detector ID of the member account for which you will delete the threat entity list, and other placeholder values that are *shown in red*.

Sample 1 starts here

Exporting generated GuardDuty findings to Amazon S3 buckets

GuardDuty retains the generated findings for a period of 90 days. GuardDuty exports the active findings to Amazon EventBridge (EventBridge). You can optionally export the generated findings to an Amazon Simple Storage Service (Amazon S3) bucket. This will help you to track the historical data of potentially suspicious activities in your account and evaluate whether the recommended remediation steps were successful.

Any new active findings that GuardDuty generates are automatically exported within about 5 minutes after the finding is generated. You can set the frequency for how often updates to the active findings are exported to EventBridge. The frequency that you select applies to the exporting of new occurrences of existing findings to EventBridge, your S3 bucket (when configured), and Detective (when integrated). For information about how GuardDuty aggregates multiple occurrences of existing findings, see [GuardDuty finding aggregation](#).

When you configure settings to export findings to an Amazon S3 bucket, GuardDuty uses AWS Key Management Service (AWS KMS) to encrypt the findings data in your S3 bucket. This requires you to add permissions to your S3 bucket and the AWS KMS key so that GuardDuty can use them to export findings in your account.

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Considerations

Before proceeding with the prerequisites and steps to export findings, consider the following key concepts:

- **Export settings are regional** – You need to configure export options in each Region where you use GuardDuty.
- **Exporting findings to Amazon S3 buckets in different AWS Regions (cross-Region)** – GuardDuty supports the following export settings:
 - Your Amazon S3 bucket or object, and AWS KMS key must belong to the same AWS Region.
 - For the findings generated in a commercial Region, you can choose to export these findings to an S3 bucket in any commercial Region. However, you can't export these findings to an S3 bucket in an opt-in Region.
 - For the findings generated in an opt-in Region, you can choose to export these findings to the same opt-in Region where they're generated or any commercial Region. However, you can't export findings from one opt-in Region to another opt-in Region.
- **Permissions to export findings** – To configure settings for exporting active findings, your S3 bucket must have permissions that allows GuardDuty to upload objects. You must also have an AWS KMS key that GuardDuty can use to encrypt findings.
- **Archived findings are not exported** – The default behavior is that the archived findings, including new instances of suppressed findings, are not exported.

When a GuardDuty finding gets generated as *Archived*, you will need to *Unarchive* it. This changes the **Filter finding status** to **Active**. GuardDuty exports the updates to the existing unarchived findings based on how you configure [Step 5 – Frequency for exporting findings](#).

- **GuardDuty administrator account can export findings generated in associated member accounts** – When you configure export findings in an administrator account, all the findings from the associated member accounts that are generated in the same Region are also exported to the same location that you configured for the administrator account. For more information, see [Understanding the relationship between GuardDuty administrator account and member accounts](#).

Step 1 – Permissions required to export findings

When you configure settings for exporting findings, you select an Amazon S3 bucket where you can store the findings and an AWS KMS key to use for data encryption. In addition to permissions for GuardDuty actions, you must also have permissions to the following actions to successfully configure settings to export findings:

- `s3:GetBucketLocation`
- `s3:PutObject`

If you need to export the findings to a specific prefix in your Amazon S3 bucket, you must also add the following permissions to the IAM role:

- `s3:GetObject`
- `s3:ListBucket`

Step 2 – Attaching policy to your KMS key

GuardDuty encrypts the findings data in your bucket by using AWS Key Management Service. To successfully configure the settings, you must first give GuardDuty permission to use a KMS key. You can grant the permissions by [attaching the policy](#) to your KMS key.

When you use a KMS key from another account, you need to apply the key policy by logging in to the AWS account that owns the key. When you configure the settings to export findings, you'll also need the key ARN from the account that owns the key.

To modify the KMS key policy for GuardDuty to encrypt your exported findings

1. Open the AWS KMS console at <https://console.aws.amazon.com/kms>.
2. To change the AWS Region, use the Region selector in the upper-right corner of the page.
3. Select an existing KMS key or perform the steps to [Create a new key](#) in the *AWS Key Management Service Developer Guide*, that you will use to encrypt the exported findings.

Note

The AWS Region of your KMS key and the Amazon S3 bucket must be the same.

You can use the same S3 bucket and KMS key pair to export the findings from any applicable Region. For more information, see [Considerations](#) for exporting findings across Regions.

4. In the **Key policy** section, choose **Edit**.

If **Switch to policy view** is displayed, choose it to display the **Key policy**, and then choose **Edit**.

5. Copy the following policy block to your KMS key policy, to grant GuardDuty permission to use your key.

```
{
  "Sid": "AllowGuardDutyKey",
  "Effect": "Allow",
  "Principal": {
    "Service": "guardduty.amazonaws.com"
  },
  "Action": "kms:GenerateDataKey",
  "Resource": "KMS key ARN",
  "Condition": {
    "StringEquals": {
      "aws:SourceAccount": "123456789012",
      "aws:SourceArn":
        "arn:aws:guardduty:Region2:123456789012:detector/SourceDetectorID"
    }
  }
}
```

6. Edit the policy by replacing the following values that are formatted in *red* in the policy example:

1. Replace *KMS key ARN* with the Amazon Resource Name (ARN) of the KMS key. To locate the key ARN, see [Finding the key ID and ARN](#) in the *AWS Key Management Service Developer Guide*.
2. Replace *123456789012* with the AWS account ID that owns the GuardDuty account exporting the findings.
3. Replace *Region2* with the AWS Region where the GuardDuty findings are generated.
4. Replace *SourceDetectorID* with the detectorID of the GuardDuty account in the specific Region where the findings generated.

To find the detectorId for your account and current Region, see the **Settings** page in the <https://console.aws.amazon.com/guardduty/> console, or run the [ListDetectors](#) API.

Note

If you're using GuardDuty in an opt-in Region, replace the value for the "Service" with the Regional endpoint for that Region. For example, if you're using GuardDuty in the Middle East (Bahrain) (me-south-1) Region, replace "Service": "guardduty.amazonaws.com" with "Service": "guardduty.me-south-1.amazonaws.com". For information about endpoints for each opt-in Region, see [GuardDuty endpoints and quotas](#).

7. If you added the policy statement before the final statement, add a comma before adding this statement. Make sure that the JSON syntax of your KMS key policy is valid.

Choose **Save**.

8. (Optional) copy the key ARN to a notepad for use in the later steps.

Step 3 – Attaching policy to Amazon S3 bucket

Add permissions to the Amazon S3 bucket to which you will export findings so that GuardDuty can upload objects to this S3 bucket. Independent of using an Amazon S3 bucket that belongs to either your account or in a different AWS account, you must add these permissions.

If at any point in time, you decide to export findings to a different S3 bucket, then to continue exporting findings, you must add permissions to that S3 bucket and configure the export findings settings again.

If you do not already have an Amazon S3 bucket where you want to export these findings, see [Creating a bucket](#) in the *Amazon S3 User Guide*.

To attach permissions to your S3 bucket policy

1. Perform the steps under [To create or edit a bucket policy](#) in the *Amazon S3 User Guide*, until the **Edit bucket policy** page appears.
2. The **example policy** shows how grant GuardDuty permission to export findings to your Amazon S3 bucket. If you change the path after you configure export findings, then you must modify the policy to grant permission to the new location.

Copy the following **example policy** and paste it into the **Bucket policy editor**.

If you added the policy statement before the final statement, add a comma before adding this statement. Make sure that the JSON syntax of your KMS key policy is valid.

S3 bucket example policy

JSON

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "Allow GetBucketLocation",
      "Effect": "Allow",
      "Principal": {
        "Service": "guardduty.amazonaws.com"
      },
      "Action": "s3:GetBucketLocation",
      "Resource": "arn:aws:s3:::amzn-s3-demo-bucket",
      "Condition": {
        "StringEquals": {
          "aws:SourceAccount": "123456789012",
          "aws:SourceArn": "arn:aws:guardduty:us-east-2:123456789012:detector/SourceDetectorID"
        }
      }
    }
  ],
  {
```

```

        "Sid": "Allow PutObject",
        "Effect": "Allow",
        "Principal": {
            "Service": "guardduty.amazonaws.com"
        },
        "Action": "s3:PutObject",
        "Resource": "arn:aws:s3:::amzn-s3-demo-bucket[optional prefix]/
    *",
        "Condition": {
            "StringEquals": {
                "aws:SourceAccount": "123456789012",
                "aws:SourceArn": "arn:aws:guardduty:us-
east-2:123456789012:detector/SourceDetectorID"
            }
        }
    },
    {
        "Sid": "Deny unencrypted object uploads",
        "Effect": "Deny",
        "Principal": {
            "Service": "guardduty.amazonaws.com"
        },
        "Action": "s3:PutObject",
        "Resource": "arn:aws:s3:::amzn-s3-demo-bucket[optional prefix]/
    *",
        "Condition": {
            "StringNotEquals": {
                "s3:x-amz-server-side-encryption": "aws:kms"
            }
        }
    },
    {
        "Sid": "Deny incorrect encryption header",
        "Effect": "Deny",
        "Principal": {
            "Service": "guardduty.amazonaws.com"
        },
        "Action": "s3:PutObject",
        "Resource": "arn:aws:s3:::amzn-s3-demo-bucket[optional prefix]/
    *",
        "Condition": {
            "StringNotEquals": {

```

```

        "s3:x-amz-server-side-encryption-aws-kms-key-id":
        "arn:aws:kms:us-east-2:111122223333:key/a1b2c3d4-5678-90ab-cdef-
        EXAMPLE11111"
    },
    {
        "Sid": "Deny non-HTTPS access",
        "Effect": "Deny",
        "Principal": "*",
        "Action": "s3:*",
        "Resource": "arn:aws:s3:::amzn-s3-demo-bucket[optional prefix]/
        *",
        "Condition": {
            "Bool": {
                "aws:SecureTransport": "false"
            }
        }
    }
]
}

```

3. Edit the policy by replacing the following values that are formatted in *red* in the policy example:

1. Replace *Amazon S3 bucket ARN* with the Amazon Resource Name (ARN) of the Amazon S3 bucket. You can find the **Bucket ARN** on the **Edit bucket policy** page in the <https://console.aws.amazon.com/s3/> console.
2. Replace *123456789012* with the AWS account ID that owns the GuardDuty account exporting the findings.
3. Replace *us-east-2* with the AWS Region where the GuardDuty findings are generated.
4. Replace *SourceDetectorID* with the detectorID of the GuardDuty account in the specific Region where the findings generated.

To find the detectorId for your account and current Region, see the **Settings** page in the <https://console.aws.amazon.com/guardduty/> console, or run the [ListDetectors](#) API.

5. Replace *[optional prefix]* part of the *S3 bucket ARN/[optional prefix]* placeholder value with an optional folder location to which you want to export the findings. For more information about the use of prefixes, see [Organizing objects using prefixes](#) in the *Amazon S3 User Guide*.

When you provide an optional folder location that doesn't exist already, GuardDuty will create that location only if the account associated with the S3 bucket is the same as the account exporting the findings. When you export findings to an S3 bucket that belongs to another account, the folder location must exist already.

6. Replace *KMS key ARN* with the Amazon Resource Name (ARN) of the KMS key associated with the encryption of the findings exported to the S3 bucket. To locate the key ARN, see [Finding the key ID and ARN](#) in the *AWS Key Management Service Developer Guide*.

Note

If you're using GuardDuty in an opt-in Region, replace the value for the "Service" with the Regional endpoint for that Region. For example, if you're using GuardDuty in the Middle East (Bahrain) (me-south-1) Region, replace "Service": "guardduty.amazonaws.com" with "Service": "guardduty.me-south-1.amazonaws.com". For information about endpoints for each opt-in Region, see [GuardDuty endpoints and quotas](#).

4. Choose **Save**.

Step 4 - Exporting findings to an S3 bucket (Console)

GuardDuty permits you to export findings to an existing bucket in another AWS account.

When creating a new S3 bucket or choosing an existing bucket in your account, you can add an optional prefix. When configuring export findings, GuardDuty creates a new folder in the S3 bucket for your findings. The prefix will be appended to the default folder structure that GuardDuty created. For example, the format of the optional prefix `/AWSLogs/123456789012/GuardDuty/Region`.

The entire path of the S3 object will be `amzn-s3-demo-bucket/prefix-name/UUID.jsonl.gz`. The UUID is randomly generated and doesn't represent the detector ID or the finding ID.

Important

The KMS key and S3 bucket must be in the same Region.

Before completing these steps, make sure you have attached the respective policies to your KMS key and existing S3 bucket.

To configure export findings

1. Open the GuardDuty console at <https://console.aws.amazon.com/guardduty/>.
2. In the navigation pane, choose **Settings**.
3. On the **Settings** page, under **Findings export options**, for **S3 bucket**, choose **Configure now** (or **Edit**, as needed).
4. For **S3 bucket ARN**, enter the **bucket ARN**. To find the bucket ARN, see [Viewing the properties for an S3 bucket](#) in the *Amazon S3 User Guide*.
5. For **KMS key ARN**, enter the **key ARN**. To locate the key ARN, see [Finding the key ID and ARN](#) in the *AWS Key Management Service Developer Guide*.
6. **Attach policies**
 - Perform the steps to attach the S3 bucket policy. For more information, see [Step 3 – Attaching policy to Amazon S3 bucket](#).
 - Perform the steps to attach the KMS key policy. For more information, see [Step 2 – Attaching policy to your KMS key](#).
7. Choose **Save**.

Step 5 – Setting frequency to export updated active findings

Configure the frequency for exporting updated active findings as appropriate for your environment. By default, updated findings are exported every 6 hours. This means that any findings that are updated after the most recent export are included in the next export. If updated findings are exported every 6 hours and the export occurs at 12:00, any finding that you update after 12:00 is exported at 18:00.

To set the frequency

1. Open the GuardDuty console at <https://console.aws.amazon.com/guardduty/>.
2. Choose **Settings**.

3. In the **Findings export options** section, choose **Frequency for updated findings**. This sets the frequency for exporting updated Active findings to both EventBridge and Amazon S3. You can choose from the following:
 - **Update EventBridge and S3 every 15 minutes**
 - **Update EventBridge and S3 every 1 hour**
 - **Update EventBridge and S3 every 6 hours (default)**
4. Choose **Save changes**.

Sample 1 ends here

Processing GuardDuty findings with Amazon EventBridge

GuardDuty automatically publishes (sends) findings as events to Amazon EventBridge (formerly Amazon CloudWatch Events), a serverless event bus service. EventBridge delivers a stream of near real-time data from applications and services to targets such as Amazon Simple Notification Service (Amazon SNS) topics, AWS Lambda functions, and Amazon Kinesis streams. For more information, see [Amazon EventBridge User Guide](#).

EventBridge enables automated monitoring and processing of GuardDuty findings by receiving [events](#). EventBridge receives events for both newly generated findings and aggregated findings, where subsequent occurrences of an existing finding are combined with the original. Every GuardDuty finding is assigned a finding ID, and GuardDuty creates an EventBridge event for every finding with a unique finding ID. For information on how aggregation works in GuardDuty, see [GuardDuty finding aggregation](#).

In addition to automated monitoring and processing, use of EventBridge enables longer-term retention of your findings data. GuardDuty stores findings for 90 days. With EventBridge, you can send findings data to your preferred storage platform and store the data for as long as you like. To retain findings for a longer duration, GuardDuty supports [Exporting generated findings to Amazon S3](#).

Topics

- [Understanding EventBridge notification frequency in GuardDuty](#)
- [Set up an Amazon SNS topic and endpoint \(Email, Slack, and Amazon Chime\)](#)
- [Using Amazon EventBridge for GuardDuty findings](#)
- [Creating an EventBridge rule for GuardDuty findings](#)
- [EventBridge rule for GuardDuty multi-account environments](#)